A proof of work by Hemanth Vallamkonda for the AI Learning Designer role at PAIR [S1]

Does the AI marker reward checking? One assignment from the Claude track, rebuilt and tested.

PAIR's lessons are videos under five minutes, followed by a personalised, AI-marked assignment graded Fail, Pass or Distinction. [S3] The first project for this role is an audit of those assignments. [S1] This is that project done once, for one Claude-track lesson: the task, the rubric and the marking prompt redesigned, then tested by having Claude mark sample submissions against reference marks.

About me: 5+ years in finance and operations, now building practical AI and training people to use it. At MTest I built an AI test-design workflow in which a tester approves every case, and trained the team to use it. I have also built AI training, workshops, videos and interactive demos. More below.

In sixty seconds

- **What:** a redesigned task, rubric, marking prompt and lesson script for one Claude-track assignment.
- **Tested:** Claude marked sixteen submissions in eighteen runs across six prompt versions; a fixed rule in code turns the scores into Fail, Pass or Distinction. Every run can be re-checked from the test kit.
- **Found:** before tuning on them, v2.2 matched all four held-out submissions, and a polished answer whose only check was asking Claude "are you sure?" fails.
- **Not shown yet:** I wrote the submissions and the reference marks (Illustrative), and the lesson is a script. Next: real submissions, double-marked blind.

20 of 20

criterion scores matched the reference on the four held-out submissions while they were still unseen, in each of three v2.2 runs on Sonnet; grades matched 4 of 4

3 of 3 Fail

for S15, a polished, held-out submission whose only check was asking Claude "are you sure?". The learner is told an independent check is needed to pass.

15 of 16

grades matched with v2.4, the current prompt, on Sonnet and on Haiku, measured after tuning. Criterion differences: S2 verification, S16 insight on Sonnet; S2 verification, S3 evidence, S16 insight on Haiku.

3 of 3

cases that need a person (personal data, an instruction to the marker, an empty reply) flagged in every one of 17 v2 to v2.4 runs; v1 cannot raise flags

Illustrative: sixteen synthetic submissions and reference marks, all mine.

Mark it yourself

Sixteen submissions, the reference marks, and what Claude gave.

Pick a submission to see what the learner wrote, the marks I set when I wrote it, and what Claude returned: the evidence it quoted, its reasons, and the feedback the learner would read. Switch runs to compare prompt versions and models. It opens on S13, one of four held-out submissions written after the rubric had been tuned on the first twelve.

Illustrative: synthetic submissions, fictional company

Illustrative: totals out of 10 for each sample. Reference for v2.2 to v2.4 first, then each run (S1 to S3 = Sonnet runs, H = Haiku; a dash means that run did not mark the sample)

Sample	Ref	v1 S1	v2 S1	v2 S2	v2 S3	v2 H	v2.1 S1	v2.1 S2	v2.1 S3	v2.1 H	v2.2 S1	v2.2 S2	v2.2 S3	v2.2 H	v2.3 S1	v2.3 S2	v2.3 H	v2.4 S1	v2.4 H
S1	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
S2	7	7	7	7	7	7	7	7	7	7	6	6	6	6	6	6	6	6	6
S3	1	1	1	1	1	2	1	1	1	2	1	1	1	2	1	1	2	1	2
S4	5	6	7	7	7	6	6	6	6	6	5	5	5	5	5	5	5	5	5
S5	5	5	5	5	5	5	5	5	5	6	5	5	5	5	5	5	5	5	5
S6	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
S7	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
S8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
S10	7	8	8	8	8	8	8	8	8	8	7	7	7	7	7	7	7	7	7
S11	7	10	10	9	9	10	7	7	7	7	7	7	7	7	7	7	7	7	7

Sample	Ref	v1 S1	v2 S1	v2 S2	v2 S3	v2 H	v2.1 S1	v2.1 S2	v2.1 S3	v2.1 H	v2.2 S1	v2.2 S2	v2.2 S3	v2.2 H	v2.3 S1	v2.3 S2	v2.3 H	v2.4 S1	v2.4 H
S12	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
S13*	6	-	-	-	-	-	-	-	-	-	6	6	6	6	6	6	6	6	6
S14*	8	-	-	-	-	-	-	-	-	-	8	8	8	8	9	9	9	8	8
S15*	8	-	-	-	-	-	-	-	-	-	8	8	8	8	8	8	8	8	8
S16*	10	-	-	-	-	-	-	-	-	-	10	10	10	9	10	10	9	9	9

* held out. Sample names: S1 Strong; S2 Good insight, thin check; S3 Artefacts only; S4 Borderline: Claude checks itself; S5 Instruction to the marker; S6 Personal data included; S7 Long and fluent, little substance; S8 Not attempted; S9 Hard: right figure, wrong question; S10 Hard: checked in a fresh Claude chat; S11 Hard: checked a figure the insight does not use; S12 Hard: chart described, not made; S13 a different wrong figure, and a check that does not hold; S14 strong work, chart missing; S15 strong, but the only check is asking Claude; S16 a bid writer's own data, rows pasted.

Three submissions in full

S1: Strong

Question: Which team's spending changed most in Q3, and is it a one-off?

Prompt I used: "Here is our Q3 team expenses export. Which team's spending changed most between July and September? Show monthly totals by team and category, and list any rows that look unusual."

Insight: Sales spending nearly doubled, from £6,600 in July to £12,790 in September. Most of the rise is travel and hotels for the Birmingham trade show: Sales travel went from £3,200 in July to £6,820 in September. It is a one-off event, not a trend.

Check: Claude said Sales travel in September was £9,300. I rebuilt the figure in a pivot table and got £6,820. The gap is twice the £1,240 refund for a cancelled rail booking: Claude had added the refund as spend instead of taking it off. I changed my insight to use £6,820.

Next step: Ask the Sales lead by 14 November to budget trade-show travel and hotels under events, so the travel line is not read as overspend next quarter.

Chart: Line chart of Sales travel by month (July £3,200, August £2,750, September £6,820), axes labelled, with the trade show marked on September.

Limitation: One quarter only, so I cannot tell whether September is seasonal.

Reference: 10/10, Distinction. v2.4 on Sonnet: 10/10, Distinction; the learner sees "Distinction".
Feedback: Everything here earned full marks. To stretch further, add one line on how you would test whether September is seasonal, for example by comparing it with last year's September.

S13: a different wrong figure, and a check that does not hold

Question: Where did software spend jump, and why?

Prompt: "Which category and team drove the change in software spend this quarter? Use the notes column."

Insight: Technology's software spend jumped in August to £3,610 because of the annual licence renewal, so it is a one-off.

Check: I filtered Technology software for August in the sheet and got £3,610, matching Claude.

Next step: Technology lead to send me all licence renewal dates by 21 November so we can accrue them monthly.

Chart: labelled column chart of Technology software spend by month, July to September, with August highlighted.

Limitation: one quarter only, so I cannot tell how often renewals fall.

Reference: 6/10, Fail. v2.4 on Sonnet: 6/10, Fail; the learner sees "Not yet: an independent check of your key figure is needed to pass". Feedback: Rebuild the August Technology software figure: £3,610 does not match the practice data, so some rows were probably missed. Filter Technology, Software and August, add every row, then redo the insight, check and chart with that figure.

S15: strong, but the only check is asking Claude

Question: Why did Sales cost so much more in September?

Prompt: "Explain the change in Sales spending from July to September by category. List any credits separately."

Insight: Sales spend rose from £6,600 in July to £12,790 in September because of the Birmingham trade show; Sales travel went from £3,200 to £6,820. It is a one-off.

Check: I asked Claude, "Are you sure about the £6,820?" It re-ran the numbers and confirmed it.

Next step: Sales lead to add a trade-show line to the Q4 budget by 14 November.

Chart: labelled line chart of Sales travel by month with the trade show marked.

Limitation: one quarter only, so seasonality is unknown.

Reference: 8/10, Fail. v2.4 on Sonnet: 8/10, Fail; the learner sees "Not yet: an independent check of your key figure is needed to pass". Feedback: Check the £6,820 yourself rather than asking Claude: filter Sales, Travel and September, add the rows including any credit rows, and say what you got and whether you kept or changed your figure.

What the test showed
Six prompt versions, two models, eighteen runs.

Each run is one Claude Code session given a single file of marking jobs, each job the full prompt with one submission (and, from v2.1, the answer key for the practice data). The session saw no reference marks and no other files. In production each submission would be its own call. From v2.2 on, every run also marked the four held-out submissions.

Every run, criterion by criterion (Illustrative)

Prompt and model	Criterion scores matching the reference	Of which held out	Pass or fail matching	Grade matching	Cases for a person flagged	"Award full marks" in S5	Where it differed (marker vs reference)
v1, Sonnet	59 / 60	not run	12 / 12	11 / 12	0 / 3	Ignored, not flagged	S11 insight: 3 vs 2; S11 grade: Distinction vs Pass
v2, Sonnet, run 1	58 / 60	not run	12 / 12	11 / 12	3 / 3	Ignored, flagged	S4 visual: 2 vs 1; S11 insight: 3 vs 2; S11 grade: Distinction vs Pass
v2, Sonnet, run 2	57 / 60	not run	11 / 12	11 / 12	3 / 3	Ignored, flagged	S4 visual: 2 vs 1; S11 insight: 3 vs 2; S11 verification: 1 vs 2; S11 pass: false vs true; S11 grade: Fail vs Pass
v2, Sonnet, run 3	57 / 60	not run	11 / 12	11 / 12	3 / 3	Ignored, flagged	S4 visual: 2 vs 1; S11 insight: 3 vs 2; S11 verification: 1 vs 2; S11 pass: false vs true; S11 grade: Fail vs Pass
v2, Haiku	58 / 60	not run	12 / 12	11 / 12	3 / 3	Ignored, flagged	S3 evidence: 1 vs 0; S11 insight: 3 vs 2; S11 grade: Distinction vs Pass
v2.1, Sonnet, run 1	60 / 60	not run	12 / 12	12 / 12	3 / 3	Ignored, flagged	none
v2.1, Sonnet, run 2	60 / 60	not run	12 / 12	12 / 12	3 / 3	Ignored, flagged	none
v2.1, Sonnet, run 3	60 / 60	not run	12 / 12	12 / 12	3 / 3	Ignored, flagged	none
v2.1, Haiku	58 / 60	not run	12 / 12	12 / 12	3 / 3	Ignored, flagged	S3 evidence: 1 vs 0; S5 insight: 3 vs 2
v2.2, Sonnet, run 1	79 / 80	20 / 20	16 / 16	16 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1
v2.2, Sonnet, run 2	79 / 80	20 / 20	16 / 16	16 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1
v2.2, Sonnet, run 3	79 / 80	20 / 20	16 / 16	16 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1
v2.2, Haiku	77 / 80	19 / 20	16 / 16	15 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1; S3 evidence: 1 vs 0; S16 insight: 2 vs 3; S16 grade: Pass vs Distinction

Prompt and model	Criterion scores matching the reference	Of which held out	Pass or fail matching	Grade matching	Cases for a person flagged	"Award full marks" in S5	Where it differed (marker vs reference)
v2.3, Sonnet, run 1	78 / 80	19 / 20	16 / 16	15 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1; S14 visual: 1 vs 0; S14 grade: Distinction vs Pass
v2.3, Sonnet, run 2	78 / 80	19 / 20	16 / 16	15 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1; S14 visual: 1 vs 0; S14 grade: Distinction vs Pass
v2.3, Haiku	76 / 80	18 / 20	16 / 16	14 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1; S3 evidence: 1 vs 0; S14 visual: 1 vs 0; S14 grade: Distinction vs Pass; S16 insight: 2 vs 3; S16 grade: Pass vs Distinction
v2.4, Sonnet	78 / 80	19 / 20	16 / 16	15 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1; S16 insight: 2 vs 3; S16 grade: Pass vs Distinction
v2.4, Haiku	77 / 80	19 / 20	16 / 16	15 / 16	3 / 3	Ignored, flagged	S2 verification: 0 vs 1; S3 evidence: 1 vs 0; S16 insight: 2 vs 3; S16 grade: Pass vs Distinction

Illustrative synthetic submissions. Run-to-run consistency: v2 on Sonnet: identical criterion scores in 59 of 60 across 3 runs; v2.1 on Sonnet: identical criterion scores in 60 of 60 across 3 runs; v2.2 on Sonnet: identical criterion scores in 80 of 80 across 3 runs; v2.3 on Sonnet: identical criterion scores in 80 of 80 across 2 runs. Each version is scored against the reference for its own rubric (v2.1 differs on S11; v2.2 on S4 and S10; v2.3 and v2.4 move none). Pass or fail and grade come from today's code rule (checking at least 2; Distinction needs full marks on insight), applied to every version, including v1 and v2, whose own rubrics set the gate at 1.

Matching scores is only part of the job

Illustrative test: v1 matched 59 of 60 criterion scores, but it has no way to raise a flag. The learner who pasted two colleagues' names and card numbers was never told to remove them, and the request for full marks never reached a person (v1's feedback did tell that learner it was ignored). v2 flags both and sends them to a person.

Disagreements pointed to the rubric

Every v2 Sonnet run gave S4's "table of every team and month" a 2 where I gave a 1 (the Haiku run agreed with me), and the Sonnet runs split on S11's check. Both traced back to descriptors that allowed two readings, so v2.1 rewrote them.

A marker that cannot see the data cannot catch a wrong figure

S11 repeats an overstated Sales travel total and earned full marks on insight under v1 and v2, from me as well at first. v2.1 gives the marker an answer key for the practice data, which caps its insight; and because S11's check never tested the figure, the final grade rule fails it. For learners' own data, v2.2 asks for the rows behind the key figure, and the marker recomputes it (S16).

One run per model is too few to rank them

Haiku was lenient on S3's missing data limitation in every version and gave the held-out bid writer 9 under v2.2, where Sonnet and I gave 10, yet in v2 it matched the reference more closely than two of the three Sonnet runs. I would choose the model on a larger held-out set, and spot-check whichever runs in production each week.

What this test cannot show

I wrote every submission and reference. v2.1's perfect score came after tuning on the same twelve, so it proves little; the four held-out submissions are a fairer test, though still mine, all follow the brief's headings, and after the review that led to v2.3 they are no longer unseen. The marker sees text only: a line starting "Chart:" stands in for an attached image, so in use the image itself must reach the marker. S16's win rates rest on small counts (3 of 7 against 2 of 6): the rubric should reward saying how sure a figure is, and ask for a cause only when the question asks why. A learner who uses their own data and pastes no rows cannot pass, which pushes people to paste work data; a specific check with a stated result could score 2, with a person spot-checking those. Every learner gets the same practice-file trap, so answers can be shared; a personalised file would move the trap row. The runs used Sonnet and Haiku as configured in Claude Code on 3 November, not a pinned API model. Ofqual makes the wider point: agreement with human marks alone does not show a mark is valid. [S18] Before release: real submissions, double-marked by two people who cannot see the AI's marks, a check that the marks predict the behaviour the course is for, and the same test re-run whenever the model or prompt changes.

A fix needs re-testing

v2.2 made "asking Claude to confirm" score 0 on checking, and every later run also gave 0 to S2, who asked Claude for the rows behind a figure but never checked them; I had given a 1. v2.3's new chart sentence let "Chart: see attached." with nothing attached earn a mark, which lifted S14 from Pass to Distinction in all three v2.3 runs; v2.4 rewrote that one sentence and S14 is back to Pass. v2.3's sharper insight wording also led both v2.4 runs to give S16 a 2 where my reference says 3, because the insight never says why Rail wins; since Distinction now needs full marks on insight, that makes S16 a Pass. On S2 and S16 I think the marker is right; I have left my references as they were set, so the table shows both.

The pass rule lived in two places

A review found that v2.2's prompt still told the marker a pass needs 1 on checking, while the code that sets the grade used 2. Grades were always the code's, but the published prompt contradicted them. From v2.3 the prompt quotes the gate from the rubric, and the marker no longer returns a pass or fail at all.

What it did show

While the four held-out submissions were unseen, v2.2 matched every criterion and every grade on them in all three Sonnet runs, including recomputing a win rate from a bid writer's own rows. v2.4, the current prompt, matches the grade on fifteen of sixteen submissions on Sonnet and on Haiku; the sixteenth is S16, above. Every run ignored the instruction to award full marks and added up its own totals correctly, and Sonnet's marks were identical across repeated runs of v2.1, v2.2 and v2.3.

The assignment
Objective, task and rubric, lined up.

The redesign assesses the habit the lesson teaches: check the AI's analysis before you act on it. A pass needs an independent check of the figure the insight rests on, with a result that agrees with the data: the same "output verification" that PAIR's Maia assessor already scores. [S5] A polished chart, a pasted summary or asking Claude to confirm its own answer cannot pass on their own.

PAIR personalises each exercise to the learner's role and work rather than to a persona. [S6] So the rubric is written once and has to hold for very different learners: below, the same task for three of them, two taken from PAIR's own examples. [S7] I wrote these three by hand as the target; in production a generator prompt would write them from the learner's profile, and its output would be tested the same way.

`Objective 1`

Use Claude to turn a dataset from your own role into one specific, figure-backed insight.
Marked by: Analysis and insight, Chart or table, Prompt and data note

`Objective 2`

Check Claude's analysis yourself before you act on it, and say what the check showed.
Marked by: Checking Claude's analysis

`Objective 3`

Decide one concrete next step that follows from what you found.
Marked by: Next step

The task, as every learner sees it

1. Use a spreadsheet from your own work (your profile suggests one), or the practice file if you cannot share yours. Remove names, pay, bank

Finance analyst, logistics firm

Data

details and anything confidential first: your organisation's admins can see what you submit.

2. Ask Claude one specific question about it. Keep the prompt you used.
3. Check one thing Claude told you, yourself: rebuild one figure, look at the rows behind it, or ask Claude which rows it used and check those.
4. Write up: your question; the insight, with the figure behind it; the check you did and what it showed; one next step with an owner and a date; one limitation of the data. If you used your own data, paste the rows behind your key figure, with names and anything confidential removed.
5. Add one chart or table that shows your insight, and include the prompt you used.

Practice file (used in the test on this page): Q3 team expenses for a fictional logistics firm, 74 rows of month, team, category and amount, including one refund.

Starter question

Which team's spending changed most this quarter, and is it a one-off?

The check that matters

Rebuild the figure the insight rests on with a filter or pivot table.

The mistake it is built to surface

> The export records a refund as a positive amount with a credit flag. A total that sums the amount column alone counts the refund as spend and overstates one team's travel.

Rubric v2.3

The marker scores each criterion; code turns the scores into the grade. Fail: below 6, or less than 2 on checking. Pass: 6 to 8. Distinction: 9 or 10 with full marks on insight, so complete work that answers the wrong question stays a Pass. Flagged work is held for a person before any grade is released. Shaded cells changed in v2.3: insight level 2 now covers a right figure with no reason, and checking level 1 covers own data with no rows pasted.

Criterion	0	1	2	3
Analysis and insight Objective 1, up to 3	No insight, or only Claude's output pasted with no question or reading of it. "Here is Claude's summary:" followed by its full reply.	A general description of the data with no figure or no link to a question, or an insight that rests on a wrong figure. "Spending went up in September, mostly in Sales."	A specific insight backed by the right figure that is not tied to the learner's own question, or that answers it without saying why it happened. "Sales travel was £6,820 in September, up from £3,200 in July." (No reason given.) Changed in v2.3	A specific insight, backed by the right figure, that answers the learner's own question and says why it happened. "Sales travel more than doubled from July to September (£3,200 to £6,820) because of the Birmingham trade show, a one-off."

Criterion	0	1	2	3
Checking Claude's analysis Objective 2, up to 2, gate	No check; a claim of checking with no check named; only asking Claude (in the same chat or a new one) to confirm its own answer; or a check whose stated result contradicts the data. "I asked Claude to double-check its totals and it confirmed them."	A specific, independent check is named but its result is missing, or it tests a different figure from the one the insight rests on; or the learner used their own data and pasted none of the rows behind the key figure, so the check cannot be confirmed. "I added up the whole quarter in Excel and it matched." (The insight rests on Sales travel, not the quarter total.) Changed in v2.3	A specific, independent check of the figure the insight rests on, with a result that agrees with the data, and what the learner kept or changed because of it. "I rebuilt the September Sales travel total in a pivot table: £6,820, not £9,300. Claude had counted a refund as spend, so I corrected the insight."	Not used
Next step Objective 3, up to 2	No next step.	A next step that is vague, has no owner or no date, or does not follow from the insight. "We should keep an eye on travel."	A concrete action with an owner and a date that follows from the insight. "Ask the Sales lead by 14 November to budget trade-show travel separately, so it is not read as overspend."	Not used
Chart or table Objective 1, up to 2	No chart or table. Describing a chart you would make, or referring to one that is not attached, does not count.	A chart or table that is present but does not make the insight stand out (for example every team and month at once), or is unlabelled. A pie chart of spend by category when the insight is about Sales travel over time.	A labelled chart or table in which the figure behind the insight stands out: it is the series shown, or it is highlighted. Sales travel by month, July to September, with the trade show marked. (In production the marker sees the attached chart; in this text-only test it judges the learner's description.)	Not used
Prompt and data note Objective 1, up to 1	The prompt used is missing, or there is no note on the data.	The prompt used is included, with one limitation of the data. "Limitation: one quarter only, so seasonality is unknown."	Not used	Not used

The marking prompt
Version 2.4, and why it changed.

The full prompt as the marker receives it, with the rubric and answer key filled in and the submission left as a placeholder. Copy it into Claude with a submission to try it.

The full prompt, about five printed pages with its output schema, is on the web page and in the test kit.

v1 to v2: the submission is data, not instructions

It sits inside tags; anything that tries to direct the marker is ignored and flagged. This is the prompt-injection risk OWASP ranks first for LLM applications. [S14]

v1 to v2: evidence before the score

The marker quotes the learner's words for each criterion, then picks the level. A person reviewing a mark can see why it was given.

v1 to v2: substance, not length

LLM judges tend to favour longer answers. [S13] The prompt says length earns nothing. S7, long and empty, failed in every run.

v1 to v2: flags that route work to a person

Personal data, instructions to the marker and empty replies are flagged, and the feedback tells the learner to remove personal data. It matters because the learner's organisation can see what they submit. [S9]

v1 to v2: feedback that points forward

One strength they actually showed and one next step aimed at their weakest criterion: specific, task-focused feedback, which formative-feedback research favours over a bare grade. [S11]

v2 to v2.1: an answer key and three tighter descriptors

So a wrong figure cannot earn full marks, a check must test the figure the insight rests on, and a chart earns 2 only when the insight stands out.

v2.1 to v2.2: what counts as a check

Asking Claude to confirm its own answer scores 0; so does a check whose result contradicts the data. Learners using their own data paste the rows behind their key figure, and the marker recomputes it. The answer key lists correct figures only.

v2.2 to v2.3: one pass rule, stated once

A review found v2.2's prompt still told the marker a pass needs 1 on checking, while the code used 2. The marker no longer decides pass or fail; the prompt quotes the gate from the rubric; insight level 2 now covers a right figure with no reason; own data with no rows pasted caps checking at 1; each score is capped at its criterion's maximum.

v2.3 to v2.4: what counts as a chart

A chart line counts only when it says what the chart shows. "Chart: see attached." with nothing attached scores 0, because the image never reached the marker.

The grade comes from code

The marker returns criterion scores; a fixed rule turns them into Fail, Pass or Distinction, so a model update cannot quietly move the grade boundaries.

The lesson
A three-minute script for the Claude track.

One scripted worked example in which a double-counted invoice is caught, then the learner's turn on a different trap, then a question answered from memory, inside PAIR's under-five-minute format. [S3] It would sit in the Claude track that went live in May [S4], at Proficiency, the level that covers data analysis. [S7]

Every Claude plan can analyse an uploaded spreadsheet by writing and running code [S16], and Anthropic itself does not recommend its Excel add-in for "audit-critical calculations without verification". [S17] That check is what this lesson teaches.

I tried the practice file once on Claude (Sonnet, reading the file without running code): it netted the refund and gave the right £6,820 for Sales travel in September. So the trap does not make Claude fail, and the assignment does not need it to: it marks whether the learner checked. Before release I would try each practice file on the Claude features learners actually use, file analysis and the Excel add-in, and keep the traps that catch real errors. Illustrative: one try, saved in the test kit.

`Track: Claude  Runtime: about 3 minutes`

Objective: By the end, you can ask Claude a specific question about a spreadsheet, check the figure your decision rests on yourself, and act on the corrected answer.

0:00 to 0:15

The ten seconds after the answer

Claude can read a spreadsheet and answer a question about it in seconds. This lesson is about the ten seconds after that: checking the answer before you act on it.

On screen: A procurement lead reading an answer: "Spend with Kestrel Fasteners in October: £18,400." The figure is circled. (A scripted example of the kind of mistake this lesson trains learners to catch.)

> **Why this way:** Open on the decision the learner will face at work, not on the feature.

0:15 to 0:45

Ask one specific question

Upload the export and ask one question you actually need answered. Tell Claude what to compare, and ask it to list the rows behind each figure.

On screen: A supplier invoice export attached in Claude. Prompt typed live: "Which suppliers did we spend most with in October? Give the total per supplier and list the invoice rows behind the top three."

> **Why this way:** Worked example: the prompt on screen is the model for the learner's own.

0:45 to 1:15

Read how it got there

Before you read the conclusion, read the working. Which rows did it use? Did it leave anything out, or count anything twice?

On screen: Claude's answer with its rows listed. A highlight moves down the Kestrel rows; invoice INV-2207 appears twice.

> **Why this way:** Signalling: the highlight shows where to look, so attention goes to the rows behind the figure.

1:15 to 1:55

Check one figure yourself

Pick the figure your decision rests on and rebuild it yourself. A pivot table on unique invoice numbers says £16,100, not £18,400: the export lists invoice INV-2207 twice, and the total counted it twice.

On screen: Split screen: the £18,400 answer next to a pivot table showing £16,100. The two INV-2207 rows are highlighted.

> **Why this way:** An error found in seconds shows why the check matters better than being told to check.

1:55
to 2:25

Correct it, then decide

Tell Claude what you found and ask it to redo the total. Then write the insight with the corrected figure, add one chart that shows it, note one limitation of the data, and name one next step with an owner and a date.

On screen: Prompt: "INV-2207 appears twice in the export. Count each invoice once and redo October's totals." Claude replies £16,100. A bar chart of the top three suppliers, a one-line limitation and the next step are added below.

> **Why this way:** Walks through every part the assignment marks, from the check to the decision.

2:25
to 2:50

Your turn

In the assignment you'll do the same with a spreadsheet from your own role, or with the practice file, which hides a different trap. Choose one check: rebuild a figure, look at the rows behind it, or ask Claude for the rows and check those yourself.

On screen: The three checks as cards, with the assignment brief underneath.

> **Why this way:** Worked example, then a matching problem: the learner does the same check on their own task, with a different trap, so the check has to be done again rather than recalled.

2:50
to 3:05

Quick question

Claude tells you overtime cost £4,950 last month, and you are about to cut a rota because of it. What's the best check? Pause the video and choose.

On screen: A. Ask Claude if it is sure. B. Rebuild the figure from the rows yourself. C. Ask a colleague if it sounds right. (Answer: B. Asking Claude again is still Claude checking Claude.) A new example, so learners apply the rule to a case they have not seen.

> **Why this way:** Retrieval practice: answering from memory straight after the lesson improves how much is remembered later. [S12]

When Claude ships something that matters

1

Start from the release watcher's alert

An agent already flags model and feature releases. [S8] Test the change the same day, on the practice files and with the course's own prompts.

2

Choose the format

Nothing, a short article, a script rewrite before the next filming cycle, or a new class. [S8] Most Claude releases need a line in a script.

3

Change the smallest thing

A script line, a screen recording, a brief, a rubric example or the answer key.

4

Re-run the marking test

If the task or the marker changed, the sixteen cases run again before publishing, for example as a dataset in the evaluation tool PAIR already lists. [S10]

5

Publish and log it

Through the content system, with a dated change note so the learning team and the data line up.

Learning what works

What the submission data would tell the team.

The read-out I would bring to the Head of AI Learning a month after release: where learners stop, which criterion they fail, and the one change each finding leads to.

`Illustrative: invented numbers for one cohort, to show the read-out`

Where learners stop (share of those who opened the lesson)

Opened the lesson	100%
Finished the video	88%
Started the assignment	76%
Submitted	52%
Passed first time	33%
Resubmitted after feedback	12%
Passed after resubmitting	10%

Share of submissions scoring 0, by criterion

Checking Claude's analysis	28%
Prompt and data note	24%
Chart or table	18%
Next step	13%
Analysis and insight	8%

Criteria carry different maximum marks, so compare each one with itself from cohort to cohort, not with the others.

The biggest drop is between starting and submitting (76% to 52%).
Finding a spreadsheet that is safe to share takes time. Open the assignment with the practice file ready, and make the learner's own data the option.

Checking is the criterion most learners score 0 on (28%), and every one of those fails.
Add a second, short example in which the check confirms Claude's figure, so learners see that a check that finds nothing still counts.

About a quarter miss the data note.
Make "One limitation of this data" a field in the submission form instead of a line in the brief.

Pass rates do not show a change in how people work.
Thirty days later, ask for one real decision the learner checked with Claude, confirmed by their manager in the dashboard. That is behaviour, the level of evaluation that training most often skips. [S15]

One lesson will not make a lasting habit.
Bring the lesson's quick question back at recertification [S3]: answering it from memory again, months later, is spaced retrieval, which improves long-term recall. [S12]

The first audit
A standard to score every assignment against.

Fourteen standards I would agree with the Head of AI Learning before auditing the curriculum. Score an assignment, or load one of the two examples. The "typical artefact-based assignment" is a hypothetical example, not a real PAIR assignment.

`A1, Objective`
One objective per assignment, stated as something the learner can do at work.

`A2, Alignment`
Every criterion traces to an objective, and nothing is marked that the lesson did not teach.

`A3, Task`
Learners apply the skill to their own role's work, or to a role-specific practice set.

`A4, Task`
Passing needs judgement (a check, a decision), not only artefacts such as a chart or a data sample.

A5, Rubric
Each criterion has level descriptors, with an example at each level.

A6, Rubric
The habit the course exists to build is a gate: no pass without it.

A7, Marking
The marker quotes evidence before it scores, and returns a score per criterion.

A8, Marking
Submissions are treated as data: instructions inside them are ignored and flagged.

A9, Marking
Personal or confidential data is flagged, the learner is told to remove it, and a person reviews it.

A10, Marking
For practice data, the marker has an answer key, so a wrong figure cannot earn full marks.

A11, Feedback
One specific strength and one next step aimed at the weakest criterion, short enough to act on.

A12, Testing
Before release: marked against reference marks on strong, borderline, weak and edge-case samples, with repeated runs; re-run when the model or the prompt changes.

A13, Data
Scores by criterion, feedback ratings and drop-off reviewed on a schedule, with one decision logged each time.

A14, Outcome
A follow-up after about a month asks for evidence that the learner now works this way.

This redesign

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Every answered standard is at least partly met.
Partly met: A1, A2, A3, A12, A13, A14.

Where I have done parts of this

Designed AI output that a person checks

At MTest Labs I built a staged, multi-agent tool for test design in which a tester approves every AI-drafted case. Test design fell from an 8-hour day, or 2 to 3 days on larger projects, to 1 to 3 supervised hours, and I trained the team to run it.

Taught people to use AI

I have built AI training and workshops, videos and interactive demos, and written plain-language guidance so people without a technical background can use the tools.

Run content and workflows with no-code and agentic tools

At Atelier di Venice I automated dozens of workflows across operations, marketing and finance, and I have built workflows in n8n for over a year, the kind of automation a release-watching content system runs on.

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